

# Day 3 - Positive, Negative, or Zero Number Check

## Question

Ek number diya gaya hai.

Check karo ki number:

- Positive hai
- Negative hai
- Ya Zero hai

## Example

### Input

15

### Output

Positive

---

### Input

-8

### Output

Negative

---

### Input

0

### Output

Zero

# Approach 1 - Using if-else if-else

## Logic

- Agar number 0 se bada hai → Positive
- Agar number 0 se chhota hai → Negative
- Warna Zero

## Dry Run

N = -10

-10 > 0 ? No

-10 < 0 ? Yes

Output = Negative

## Time Complexity

O(1)

## Space Complexity

O(1)

## Java Code

```
public class Day3 {  
    public static void main(String[] args) {  
  
        int n = -10;  
  
        if(n > 0) {  
            System.out.println("Positive");  
        }  
        else if(n < 0) {  
            System.out.println("Negative");  
        }  
        else {  
            System.out.println("Zero");  
        }  
    }  
}
```

# Approach 2 - Using Nested if

## Logic

Pehle check karo number zero hai ya nahi.

Agar zero nahi hai tab positive ya negative check karo.

## Java Code

```
public class Day3 {  
    public static void main(String[] args) {  
  
        int n = 25;  
  
        if(n != 0) {  
  
            if(n > 0)  
                System.out.println("Positive");  
            else  
                System.out.println("Negative");  
  
        } else {  
            System.out.println("Zero");  
        }  
    }  
}
```

## Time Complexity

$O(1)$

## Space Complexity

$O(1)$

# Approach 3 - Using Ternary Operator

## Java Code

```
public class Day3 {  
    public static void main(String[] args) {  
  
        int n = -5;  
  
        String result = (n > 0)  
            ? "Positive"  
            : (n < 0)  
            ? "Negative"  
            : "Zero";  
  
        System.out.println(result);  
    }  
}
```

## Time Complexity

$O(1)$

## Space Complexity

$O(1)$

# Interview Follow-Up Questions

## Q1. Kya Zero Positive hota hai?

**Answer:**

Nahi.

```
Zero na Positive hota hai  
Zero na Negative hota hai
```

---

## Q2. Integer Range kya hoti hai Java me?

```
Minimum = -2147483648
```

```
Maximum = 2147483647
```

---

## Q3. Floating Point Number ke liye bhi ye logic chalega?

Haan.

```
double n = -12.5;
```

Same comparison operators kaam karenge.

---

# Real Life Use Cases

## Banking System

```
Balance > 0 → Positive Balance  
Balance < 0 → Loan / Debt  
Balance = 0 → Empty Account
```

## Temperature Monitoring

```
Temperature > 0  
Temperature < 0  
Temperature = 0
```

---

# Summary

## Approach Time Space

if-else if  $O(1)$   $O(1)$

Nested if  $O(1)$   $O(1)$

Ternary  $O(1)$   $O(1)$

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# Interview Concept of the Day

## Difference Between if and else if

```
if(a > 0)
```

Ye independent condition hai.

```
else if(a < 0)
```

Ye tabhi check hogi jab pehli condition false ho.

Isliye if - else if - else chain efficient hoti hai.

# Practice Questions

Predict Output:

**1**

```
int n = 100;
```

**2**

```
int n = -999;
```

**3**

```
int n = 0;
```

**4**

```
int n = 2147483647;
```